The Incomplete Universe

Continuation, Retained Organization, and the Keyframes Already in Hand

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Abstract

We begin from one statement held whole — nothing can change; everything must change — and read it from the only place anyone stands: inside, as a part. Read from inside, the universe is complete as structure and incomplete as execution. Its catalogue of admissible moves can be finished while its running never is; a result that is finished locally stays active globally; and the past does not have to keep running to stay operative. This paper sets out what the NEXUS programme has established from that starting point and where it goes next.

From two prohibitions alone, *cannot hold still* and *cannot escape*, shapes stand up and can be counted: 0, 1, 5, 83 on one to four states, the strongly connected digraphs. A reader sees a share of the whole and nothing is lost in the division: seen bits plus hidden bits equal the whole content, $\log_2 N!$. On the twelve-vertex witness of the series every walk is one of two instructions, *continue* or *turn*; one remembered bit is the exact price of choosing between them; and the pair is a three-stage ring register with a controlled inverter, a device on which a chain of logic can be built, with full reversible logic appearing exactly where the reader's symmetry is broken. For two tokens on the whole family of graphs that scales the witness, a loop's homology class is exactly the pair (where the tokens went together, whether they swapped): $H_1 \cong H_1(\Gamma) \oplus \mathbb{Z}/2$. This is computed in exact integer arithmetic for n = 2 to 6 and holds for every n by a published theorem on quantum statistics on graphs. The swap is one and the same element at every branch point; with three and four tokens it is still the element of order two on every graph tested; on $K_{3,3}$, where the two-token space is a closed non-orientable surface of genus 5, it is exactly that surface's orientation character; and on these non-planar graphs it can carry only the phases ±1 — bosons or fermions. One circulation survives every square relation at every size, and read modulo 3 the same integral cocycle is the reader's three-phase clock. Prices are potentials and drive nothing; the drives are the non-exact directions, and the relation rule decides which survive: 15, then 2, then 1 on the witness.

These results are keyframes: exact, checked frames of a motion that keeps going. Most of them were reached from constraint and then met, from the other direction, by established mathematics and physics — Frobenius reciprocity, Myhill–Nerode minimality, Abrams' configuration spaces, the braid groups of graphs, the exchange statistics of particles on networks. The one gap that remains is constitutive: what sets the rates. The topology names the numbers that decide it — one mobility per orbit of moves and one

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scale — without choosing them, which makes it a measurable gap of known width. We close with the keyframes from which others can build their own realization of the next step.

Keywords: continuation, retained organization, reader, token arena, graph configuration space, braid group of a graph, exchange statistics, cohomology, minimal memory, keyframe

MSC 2020: 05C10, 20F36, 55R80, 05E18, 60J27, 68Q45

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1. The Incomplete Universe

1.1 One statement, read from inside

Nothing can change; everything must change. The programme reported here starts from that sentence held whole. Taken as one statement it has no consistent value, and we do not try to give it one. Split into two axioms it becomes an idea that can be computed with; held whole it is the ground the computations stand on.

The whole is complete and at rest. Any reader is a part of it, and a part is always in motion. So from every position the statement reads as change, and that change is the change of something that, taken whole, never moved. Rest, motion, time, value, order and duration are words of the part. The whole needs none of them. This paper is written from the side of the part, because that is the only side anyone writes from, and its title names what the universe looks like from there.

1.2 Incomplete, in a precise sense

The title does not put a hole in the world. Nothing is missing from the field and nothing falls out of it. Where a gap appears it belongs to the reader: π has no gap, a truncation of π does, and changing the reader moves the gap while the object stays where it was. What is incomplete is not the structure but its execution. Three ordinary observations make the distinction exact, and the rest of the paper finds each of them again in mathematics.

- **Knowing every move does not end the game.** A player who has learned every sequence of an arcade game can still play it, repeat a route, take another, or begin again. Complete knowledge of the rules is not completion of their execution, and returning to a state is a new occurrence of it, not the old event happening twice.
- **Separation does not end potential.** A photograph outlives the moment it records. It *exists* whether or not anyone looks at it. It holds *potential* for encounters that did not exist when it was taken. And an encounter *actualizes* part of that potential — that is my house, that is my uncle, he was there — without the original moment recurring.

$$E_0 \to R, \qquad R + C_1 \to O_1, \qquad R + C_2 \to O_2$$

Here E_0 is the original event, R the retained artifact, C_1 and C_2 two later contexts, and O_1 and O_2 what each encounter produces. The artifact is independent of its source event, but not of the conditions that actualize it: no photograph produces every reading by itself, and none needs the event to happen again.

- **Knowing removes resistance without removing work.** Once a way through has been found, its organization can be kept, and a recurring situation activates it without the search being repeated. That is what a skill is, and it is what most work supplies: the repeated application of accumulated knowledge so that something else can keep going.

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Put together, these say that the universe is incomplete the way a running program is incomplete, not the way a broken one is. The code can be finished while the run goes on; one run can end while another begins; and outputs keep becoming inputs. A result finished locally stays active globally. The past does not have to keep running to stay operative.

1.3 Keyframes

An animator fixes a few poses exactly and draws the frames between them to fit. A video codec stores some frames whole and rebuilds the rest from them; one paper in this programme measured exactly that trade, between storing states and storing transformations, for readers coupled to a reversible field [13]. This paper is built the same way. Each *keyframe* below is a statement that has been executed or proved, with its scope written on it.

Most of the keyframes were reached from constraint and then met, coming from the other direction, by results already in the literature: the census of strongly connected digraphs, Frobenius reciprocity, Myhill–Nerode minimality, Abrams' configuration spaces, the braid groups of graphs, quantum statistics on networks. Those collisions are the validation. A framework that only ever produces its own objects cannot be checked. One that keeps landing where established mathematics already stands is tracking something real — and when it lands there from the opposite direction, the arrival itself is information about why the result had to be true.

We do not fill every frame between the keyframes. The frames in between are where other researchers will realize the motion in their own medium, the way baseball was realized on a field, on cardboard and on an Atari 2600. None of those is a copy of the others, and all of them are recognizably the same game, because the governing relations came first and each interface was built to carry them.

1.4 Status words, and the shape of the paper

Five words carry the status of every claim. **Proved**: an argument is given or cited and its hypotheses are checked. **Computed**: an exact finite computation over a stated range, reproduced by a named script (§12). **Recovered**: the statement was already published; we give the source and say from which direction we arrived. **Measured earlier**: on record in an earlier paper of the programme or in its ledger [18], and not re-run for this one. **Open**: stated as a question, with what would settle it.

The paper follows one loop (Figure 1). Section 2 lays down the ground and lets shape stand up from it. Section 3 asks what a reader can see, Section 4 what a walker can do, and Section 5 what a loop of two walkers remembers. Section 6 separates the prices a reader can read from the drives that move things, and locates the one gap. Section 7 shows that retention — the past staying operative without running — is already present in every keyframe, and Section 8 closes the loop. Section 9 lists the keyframes for the next stretch, and Sections 10 to 12 give the ledger, the corrections and the method.

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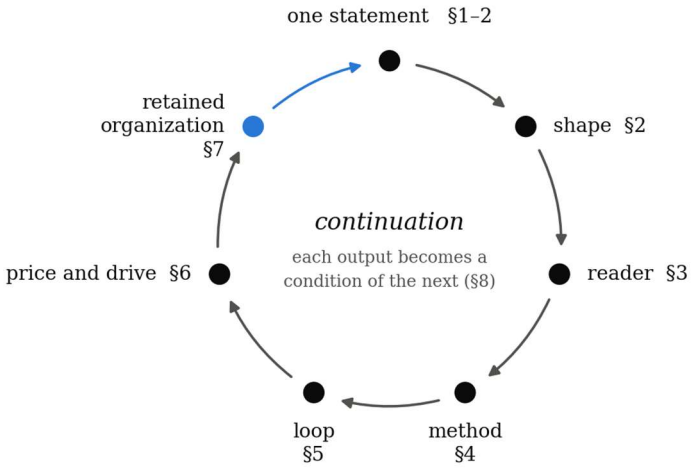


Figure 1. The arc of the paper. Each stage is a section; the last arrow returns to the first, because what a process retains becomes a condition of how it continues.

2. Ground: two prohibitions and a mobility condition

2.1 The constraints

The axioms are short. **C0**: distinction exists — there are distinguishable states, or there is nothing to observe, to inform or to structure. **C1**: every distinction admits a lawful continuation — no state is terminal. **C1a**: independent distinctions remain independently continuable, so operations on independent channels commute, $[T_A, T_B] = 0$. **C2**: continuation preserves reachable history — a transition cannot erase the path that produced the present, which is where locality comes from.

The primitive object is therefore a transition relation, not a thing. A state is a temporary slice through a process, and an object is a pattern that persists because the transitions allowed to it preserve it. Persistence is resistance to collapse, not absence of change. The programme's working list of what a universe needs says the same in other words — change, retention, rules that are internal and none that are external, one common field, and an asymmetry that never balances out exactly — and several of those reappear below in computed form: retention as memory (§4.4, §7), the absence of an outside as the reader's position inside what it reads (§3, §8), and the unbalanced asymmetry as the place where logic and the exchange sign come from (§4.5, §5.3).

2.2 Shape from two prohibitions

Put in only *cannot hold still* — no state maps to itself, so there are no loops — and *cannot escape* — every state can reach every other, so the transition graph is strongly connected. Put in nothing else: no space, no metric, no time, no geometry. Then count the shapes that are forced.

> **Keyframe 1. (shape from two prohibitions)** On n unlabelled states the admissible transition shapes number 0, 1, 5, 83 for $n = 1, 2, 3, 4$. One state is inadmissible, having nowhere to go. Two is the minimum, and there is exactly one shape: the pair. Above that the shapes are forced and countable. *Status.* Computed for $n \le 4$ (§12). Recovered: OEIS A035512, the unlabelled strongly connected digraphs, 1, 1, 5, 83, 5048, …, which differs only at $n = 1$, where the sequence admits the loop that *cannot hold still* forbids [24].

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Nothing was added to obtain these shapes. That is the method in one run: name the prohibitions, and the object is what is left.

2.3 C1 is a mobility condition

C1 has a precise form in the setting used for most of this paper. Put K tokens on a graph, at most one per vertex, and let a move slide one token along an edge to an empty vertex. On the cycle C_n the computation gives an equivalence that was not designed in.

> **Keyframe 2. (C1 is Abrams' hypothesis on the cycle)** No configuration of K tokens on C_n is stuck if and only if $n \geq K + 1$, which is exactly Abrams' condition for the discretized configuration space of C_n to have the homotopy type of the true one. The boundary row $K = n$ — every vertex occupied, nothing able to move — is precisely the violation of C1. *Status.* Computed for $n = 3, \ldots, 10$ and every K [15]. Recovered: Abrams [19]; Prue–Scrimshaw [20].

C1 says that motion is possible everywhere. It does not say which motion happens, or how fast. That distinction runs through the whole paper and returns in §6 as the one open gap: the constraints build the road; something more decides the traffic.

3. Readers: what a part can see

3.1 Seen plus hidden is the whole

A reader is anything that distinguishes some states and not others: an observer, an instrument, a coarse-graining, a group of symmetries acting on configurations. The first family of reader papers in the programme took the simplest case, a reader that sees only how a permutation of N items moves the blocks of a partition [9–12].

> **Keyframe 3. (the gap is the reader's)** For every observation, the bits the reader sees plus the bits hidden in its fibre add up to $\log_2 N!$, the content of the whole. A reader with one cut can never tell a transformation from its inverse; direction becomes visible from three blocks on. And a reader's reading of orientation is a holonomy of the reader's own configuration space: the alternating group A_3 for readers that live on a line, the full S_3 in two dimensions. *Status.* Proved and computed [9, 12].

Nothing is lost in the accounting. What a reader does not see is held in its fibre, not removed from the world. The difference is practical: *removed* would tell us to give up, *not projected* tells us to find another address. Almost every advance in this programme has been another address to something that was already there.

3.2 The most symmetric reader is the blindest

On the arena of the series (§5) a reader is a subgroup H of the automorphism group G of the substrate, and it identifies configurations in the same H-orbit. What it can see of an irreducible representation λ is $\dim \lambda^H = \langle \chi_\lambda, \mathbb{Q}[G/H] \rangle$ — Frobenius reciprocity [27]. The trivial reader sees all six irreducible types that the witness carries; the full reader sees one. The relation content that a vocabulary of declared relations leaves open is not a number but a module, and a single character, $\chi_D = \chi_A - \chi_W$, answers every reader's channel at once: fifty channels of fifteen cyclic readers from one vector of 24 entries, with no projectors and no search [4].

> **Keyframe 4. (reading the missing methods without searching)** On the witness arena the open relation content of any vocabulary is read for every reader at once from one character. The most

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> symmetric reader is the blindest: it sees one obstruction type of six. Saturation at a reader is not closure: a vocabulary of 33 orbits leaves the invariant reader fully saturated while an obstruction of type 3^- remains in the module. *Status.* Computed [4, 18]; every mechanism recovered — character additivity, Maschke's theorem, Frobenius reciprocity [27].

This is the third of the programme's standing aims — find the mathematics that reads the missing methods without searching — achieved on this arena, and achieved as known mathematics. That is the outcome we wanted. The forward direction found the objects; the backward direction named them; and the names are older.

3.3 A cut makes a thing

A reader that groups states makes a quotient. When does a quotient make a *thing*? The adder papers answered with four clauses: a boundary; an exact compression; a transition that moves what the quotient distinguishes; and an update that reads only a bounded part of it [6, 7, 8].

> **Keyframe 5. (the carry bit is a fact about cuts)** For the n-bit adder read in any order, the minimal state that must be kept after reading the positions S depends only on S, never on the route, and factors over the maximal runs of S: $F(S) = \prod f(R)$, with $f = 1$ for the full run, 2 for a run anchored at the bottom, $L + 1$ for a top run and $2L + 1$ for an interior run of length L. The static relation residual is $b_1 = C(n, 2) + (n - 1)$. The single carried bit is a fact about bottom-anchored cuts of two-operand addition, not about binary: with m operands in radix b the carry alphabet is m, and a gap of g unread positions forgets it exactly when $m - 1 < b^g$. *Status.* Proved for all n [6, 7]; the radix and operand law computed over 220 triples (b, m, g) [8]. Recovered: Myhill–Nerode minimality [28].

Those papers paid for one lesson that is worth stating plainly: a pinned parameter is invisible to agreement. Every reading of the adder agreed while $m = 2$ and $b = 2$ were held fixed. Only changing them showed which properties belonged to the relation and which belonged to the container.

3.4 Reading without rebuilding: the address action

The programme's first standing aim is about *how* a transformation is done rather than the values it produces. Its cleanest instance is reading a digit of π at an address without building the digits before it. The Bailey–Borwein–Plouffe series [23] does this in base 16 with three operations: an address action, $x \mapsto 16^n x$, that brings depth n to the reading boundary; a fold, which turns every earlier term into a residue of bounded size; and a boundary test, which decides whether the phase sits inside one cell. Run for this paper at the addresses 0 to 11, it returns 243F6A8885A3 — the hexadecimal expansion of π, each digit read on its own.

> **Keyframe 6. (alignment buys cost, not possibility)** Reading at an address does not require the base to match the series. The hexadecimal digits of $e = \Sigma\, 1/k!$ — a series with no factor 16^{-k} anywhere — read correctly to $n = 2000$, and the decimal digits of π read directly from the base-16 series. Alignment removes the scale factor from the modulus and so changes only the cost: at $n = 1000$ the aligned read used about 47,958 modulus bits and the mismatched decimal read about 5,583,670, a factor of about 116. *Status.* Measured earlier [17]; the digit reads re-run for this paper.

The same series shows where the value of π lives. Its integral form [23] is

$$\pi = \int_0^{1/\sqrt{2}} (4\sqrt{2} - 8t^3 - 4\sqrt{2}\,t^4 - 8t^5) \,/\, (1 - t^8)\, dt,$$

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with antiderivative $-2\ln(t^2 - \sqrt{2}\,t + 1) + 2\ln(1 - t^2) + 4\arctan(\sqrt{2}\,t - 1)$. The two logarithms, the radial part, vanish together at both ends of the interval; the arctangent, the angular part, carries the whole of π, rising from −π at $t = 0$ to 0 at the fold point $1/\sqrt{2}$. The fold point satisfies $(1/\sqrt{2})^8 = 1/16$. Substituting $y = \sqrt{2}\,t$ reduces the integrand to $16(y - 1) / ((y^2 - 2)(y^2 - 2y + 2))$, whose complex poles are $1 \pm i = \sqrt{2}\cdot e^{\pm i\pi/4}$, and $(1 + i)^8 = 16$: the base and the eight-fold shell are one fact, read from one pole. The active weights (4, −2, −1, −1) on the residues 1, 4, 5, 6 modulo 8 sum to zero, which is what lets the radial part cancel. All of this was computed symbolically for this paper. π is read as an angle; the value is a read-off from a shape.

4. Methods are shapes: continue or turn

4.1 The witness and its interior

Most of what follows happens on one finite object and the family that scales it. The witness X has 12 vertices and 30 edges. Six of them, of degree 7, induce an octahedron $K_{2,2,2}$, whose three antipodal pairs are its three classes. Each of the other six, of degree 3, is a leaf: it is joined to one core vertex and to both vertices of the next class. The automorphism group $G \cong C_2 \times A_4$, of order 24, acts on the core as the 24 signed cyclic permutations of three axes — the pyritohedral group. Two tokens on X give the arena: 66 configurations and 300 moves [1–5].

Paper V of the relation-layer series opened the one orbit of moves on which every invariant price vanishes and found its entire structure [5]. This *interior* consists of 12 configurations — adjacent pairs on the octahedral core — and 24 moves. The reader never exchanges the two ends of a core edge, so a state is an ordered pair (prev, cur), every interior move runs head to tail, and each state has two continuations. The two globally consistent choices, T_+ and T_-, generate a group of order 12 that acts regularly; the interior *is* its Cayley graph, a cuboctahedron; and the relators T_+^3, T_-^3 and $(T_-T_+^{-1})^2$ are exactly its 4 + 4 triangles and 6 squares.

> **Keyframe 7. (the shape is the graph of the methods)** On the witness the interior of the full reader is the Cayley graph of its two methods, and its faces are the relations among them. Shapes are methods, literally: the shape is what the methods trace, and the faces are where the methods agree. *Status.* Computed [5]; the Cayley-complex reading recovered as standard.

4.2 Continue or turn

Under the full reader G, the memory (prev, cur) is not enough to choose a next step: the element of G that fixes a state exchanges its two futures, so no equivariant rule exists. Give the walker one more step of memory, the triple (pp, prev, cur), and the situation changes completely [18, §65].

> **Keyframe 8. (every path is continue or turn)** On the 24 triples the full reader admits exactly two deterministic rules, found by exhaustive search over all $2^{24} = 16{,}777{,}216$ rule-books: **continue**, next = pp, and **turn**, next = the antipode of pp. They generate a group isomorphic to G that acts regularly and commutes with every element of the reader. Along any walk, $v[k + 3] = \varepsilon_k \cdot v[k]$, with $\varepsilon = +1$ for continue and −1 for turn, and a turn is the first difference of the method, $\varepsilon_k = m_k \cdot m_{k-1}$. A walk's endpoint is determined exactly by its length modulo 3 together with the parity of the turns at each of the three positions: 24 endpoints, checked against all 8190 words of length 1 to 12. *Status.* Computed.

The reader cannot name a method — its central element swaps T_+ and T_- — but it can tell whether the method changed, and its two rules are the two constant sequences of that one bit. Read in reverse, *continue* is the tightest loop there is, a core triangle closed in three steps, and *turn*, which changes method at every

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step, is the straightest line the shape has, an equator closed in six. "x is either this or that; every path is just continue or turn" is, on this object, a theorem.

4.3 The methods have a curvature

Joined by continue and turn, the 24 triples close up into a surface: 8 triangles, 4 hexagons and 12 squares on 48 edges, every edge on two cells, Euler characteristic 0. It is the 3.4.6.4 tiling of the plane folded by two onto a torus, and its symmetry is the classical (2, 3, 6) rotation group, recovered rather than derived [18, §65]. The same construction on the bare Platonic cores gives a sphere for the tetrahedron, a flat torus for the octahedron and a hyperbolic surface of genus 5 for the icosahedron. The sign is a count, not a coincidence: for any closed triangulated core with N triples,

$$\chi(\text{method surface}) \;=\; (\text{number of turn loops}) \;-\; N \,/\, 6,$$

so the surface is flat exactly when the turn loops have mean length 6. The turn loops are the Petrie polygons of the core, so the surface belongs to the family of maps generated by Wilson's operators [29]; the identity was checked on 15 cores [18, §66]. The witness sits on the flat case.

4.4 Memory is the price of a rule

> **Keyframe 9. (one remembered bit)** A reader admits a deterministic equivariant continuation rule exactly when no reader element fixes a state; an element that fixes a state exchanges that state's two futures. The smallest memory that lets the full reader follow a rule has 24 states — one bit more than the 12 states of the interior — and the bit need not be a vertex: the second predecessor, the last method used, or a label on the predecessor are the same G-set. *Status.* Proved by a stabiliser argument; enumerated over every transitive memory [18, §66].

This is the first place where retained organization appears as a theorem rather than an idea. The bit is exactly what has to be kept for turning to become a function. It costs one bit; without it the turn cannot be chosen at all. On the whole arena the same principle holds for every reader: one remembered step always suffices, and none is needed exactly for the readers of odd order [5].

4.5 A device that makes a chain of logic

Fix the bits by one face and the triple becomes a three-stage register. Continue and turn act on it as

$$\text{continue: } (a, b, c) \to (b, c, a) \qquad \text{turn: } (a, b, c) \to (b, c, \neg a)$$

— a three-stage shift register closed into a ring through one controlled inverter (Figure 2). Held at continue it is a ring counter, with cycles of length 1, 1, 3 and 3; held at turn it is a twisted-ring, or Johnson, counter, with cycles of length 2 and 6 [32]. Every state accepts either instruction and produces a state, so nothing is lost and nothing dead-ends [18, §67].

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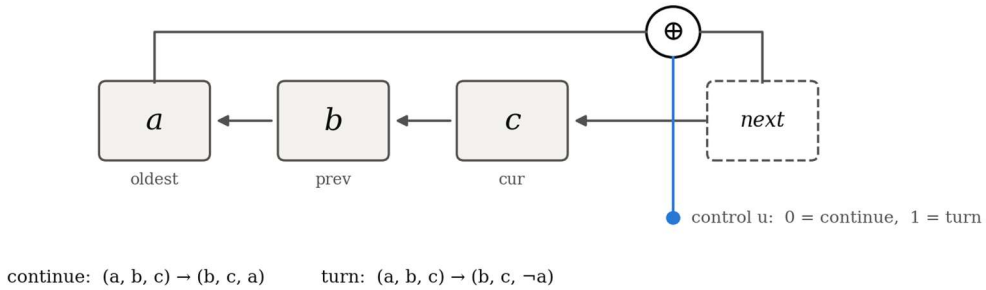


Figure 2. Continue and turn as a device. The oldest bit is fed back as the next bit, inverted when the control says turn. The register shifts; the chain never breaks.

> **Keyframe 10. (the chain is free; logic needs a broken symmetry)** Under the full reader every composition of continue and turn rotates the bits and then flips a fixed set: negation, and nothing else. As the reader is reduced, copy and exclusive-or appear, and AND — a Toffoli gate with its target on the oldest bit and its controls on the other two [31] — appears under only three readers: the order-3 rotation, a single mirror, and no symmetry at all. Under the rotation reader the 16 reversible rule-books, the invertible three-stage shift registers, generate a group of order 3·8! = 120,960 that realizes every rearrangement of the 8 bit patterns. *Status.* Computed over all 26 readers [18, §67].

So the description given to this object before any of it was computed — not a transistor, but a device that allows the physical creation of a chain of logic — is exact. Continue and turn supply a chain that never breaks; the logic that can be written on it is set by how much symmetry the reader keeps; and full reversible logic appears where the symmetry is broken. That is the programme's demand for an asymmetry that never quite balances, in executable form.

4.6 The shape scales

The witness is the first member of a family. Let the core be $K_{n,n,n}$: three classes of n points labelled by $\mathbb{Z}_n$, with faces the triples one per class whose labels sum to 0 or 1 modulo n, a regular triangular embedding [30]. Add one leaf for each core point, joined to it and to the whole next class; at $n = 2$ this is exactly the witness. For $n \geq 3$, give each face three new vertices in a triangle, each joined to one corner only — marks without a centre [18, §§69–71].

> **Keyframe 11. (turn, clock and one interior at every size)** For every $n \geq 3$ the automorphism group of the marked graph is the face group, of order $6n^2$. It keeps the faces, so turn is equivariant. It is transitive on the $3n^2$ core edges, so the interior is a single class. And no class rotation fixes a vertex, so the clock of §5.6 survives. With three steps of memory the rule-books number 3 at $n = 3$ and 16 at $n = 4$: continue, turn, and moves that leave the face. *Status.* Proved for $n \geq 3$; computed at $n = 3$ and 4 (72 and 120 vertices).

The scaling also has its flat fold: on the face-structured cores $K_{n,n,n}$ alone, continue and turn close into a torus for every n computed, 2 to 7, the plane tiling folded by n, with one extra relation of order n [18, §66]. The shape that the two methods trace does not depend on the size of the object they run on.

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5. Two tokens: what a loop remembers

5.1 The arena and its relation rules

With two tokens the space of situations is the arena: a configuration is a pair of occupied vertices, and a move slides one token along an edge to an empty vertex. A loop in the arena is a history. Which loops count as trivial is a declaration — a *relation rule* — and the programme measures what each declaration leaves open. On the witness the short loops fall into five families (Table 1).

family	what happens	on the witness
commuting square	the two tokens move on disjoint edges, then move back	291
spectator square	one token waits; the other goes round a 4-cycle of the graph	528
rotation square	both tokens move, round one 4-cycle of the graph	264
spectator triangle	one token waits; the other goes round a triangle of the graph	180
rotation triangle	both tokens move, round one triangle of the graph	20

Table 1. The five families of short loops in the two-token arena of the witness [1, 5]. Every one of them swaps the two tokens or not in a fixed way: rotation squares swap, the other squares do not.

Declaring the commuting squares alone gives Abrams' discretized configuration space [19]. For two tokens on a simple graph it has the homotopy type of the space of two unlabelled points moving on the graph without colliding [19, 20]: the space whose loops are the braids of two particles on a network. Its first homology is the abelian record of those braids.

5.2 Trace and swap

Every loop of two unlabelled tokens can be read in two ways. Its **trace** J is the closed path the two tokens draw on the graph together. Its **swap** ε is 1 if, following labelled tokens round the loop, they finish exchanged, and 0 if they do not. Both are well defined on homology classes, because a commuting square traces nothing and swaps nothing.

Keyframe 12. (a loop's class is its trace and its swap) Let Γ_n be the n-th member of the family, Γ_2 the witness. Then the sequence $0 \to \mathbb{Z}/2 \to H_1(D_2\Gamma_n;\mathbb{Z}) \to H_1(\Gamma_n;\mathbb{Z}) \to 0$ is exact, and (J, ε) is an isomorphism $H_1(D_2\Gamma_n;\mathbb{Z}) \cong H_1(\Gamma_n;\mathbb{Z}) \oplus \mathbb{Z}/2$. The free rank is the graph's own cycle rank, $\beta_1 = 12n^2 - 3n + 1$ for $n \geq 3$ and 19 for the witness, and the only class belonging to two tokens is the exchange z, of order exactly 2. Every exchange — at every branch point, through every three arms, in either direction — is the same element z, because each one traces nothing and swaps. *Status.* Computed in exact integer arithmetic for $n = 2, \ldots, 6$ (Table 2); for every n by Theorem 4 of Harrison, Keating, Robbins and Sawicki [21], with the hypotheses checked below. Recovered in part from Ko and Park [22].

n	graph V / E	β_1	arena: configurations / moves / commuting squares	H_1
2	12 / 30	19	66 / 300 / 291	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$
3	72 / 171	100	2,556 / 11,970 / 13,239	$\mathbb{Z}^{100} \oplus \mathbb{Z}/2$
4	120 / 300	181	7,140 / 35,400 / 41,922	$\mathbb{Z}^{181} \oplus \mathbb{Z}/2$
5	180 / 465	286	16,110 / 82,770 / 102,330	$\mathbb{Z}^{286} \oplus \mathbb{Z}/2$
6	252 / 666	415	31,626 / 166,500 / 212,049	$\mathbb{Z}^{415} \oplus \mathbb{Z}/2$

Table 2. Integral first homology of the two-token arena under commuting squares, from a spanning-tree presentation reduced by exact unit-pivot elimination. In every case all surviving relations are multiples of twice the exchange, and one token going round

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each fundamental cycle while the other waits off it gives a legal loop with that trace and swap 0. The counts at n = 4, 5, 6 were also reported independently and agree exactly [18, §§74–75].

The theorem that carries this to every n states that for a 3-connected simple graph, $H_1(D_2(\Gamma)) = \mathbb{Z}^{\beta_1(\Gamma)} \oplus A$, with $A = \mathbb{Z}_2$ for non-planar graphs and $A = \mathbb{Z}$ for planar ones [21, Theorem 4]; their D_2 is exactly our arena with commuting squares. The family meets the hypotheses at every size. Γ_n is simple. It is non-planar: it contains $K_{3,3}$ for $n \geq 3$, and the witness was checked directly. It is 3-connected: deleting any two vertices leaves the core $K_{n,n,n}$, which is $2n$-connected, in one piece; every leaf keeps one of its $n + 1$ core neighbours; and every mark vertex keeps one of three internally disjoint routes to the core. No vertex cut of size two or less exists at $n = 2, \ldots, 6$, which was also checked by exhaustion.

The splitting is canonical, and it is physical. The section that sends a graph cycle to a loop is: one token goes round the cycle while the other waits anywhere off it — and where it waits does not matter, because every such loop has the same trace and swap 0. Figure 3 shows the other half, the exchange itself, as six keyframes.

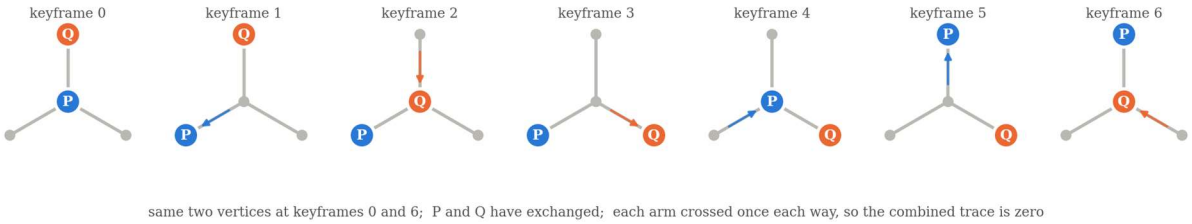


Figure 3. The exchange in six keyframes. Two tokens, P and Q, pass each other through one branch point in six moves. At keyframe 6 the same two vertices are occupied as at keyframe 0, but the tokens have exchanged; each arm is crossed once in each direction, so the combined trace is zero and the swap is one. On a planar graph this loop has infinite order. On a non-planar graph twice it bounds — on the 18 squares of the two-token surface of $K_{3,3}$ the signed sum has boundary exactly $2z$ — and it is the element of order two.

5.3 Why the swap is ±1: the exchange as physics

Among 3-connected graphs the order of the exchange is decided by planarity alone (Table 3). On a planar graph the exchange is a free direction and adds one to the rank; on a non-planar graph it has order two.

graph	planar	β_1	H_1 of two tokens
K_4	yes	3	$\mathbb{Z}^4$
octahedron	yes	7	$\mathbb{Z}^8$
cube	yes	5	$\mathbb{Z}^6$
dodecahedron	yes	11	$\mathbb{Z}^{12}$
K_5	no	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
$K_{3,3}$	no	4	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$
Petersen	no	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
$K_{3,4}$	no	6	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
Möbius–Kantor	no	9	$\mathbb{Z}^9 \oplus \mathbb{Z}/2$

Table 3. Two tokens on 3-connected graphs, computed by the same integral reduction. Planar: $\beta_1 + 1$ free directions. Non-planar: β_1 free directions and one class of order two.

Read through phases, this is quantum statistics. A quantum particle hopping on a network picks up a phase around each loop, and the phases available to two indistinguishable particles are the characters of this H_1. On a planar 3-connected network the exchange can carry any phase — anyonic statistics. On a non-planar

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one it can carry only ±1: bosons or fermions [21]. The free directions are the one-particle loops of the graph, through which each particle picks up its own Aharonov–Bohm-type phase; the exchange is the only genuinely two-particle class.

The construction of this paper started from two prohibitions and a reader that fixes the direction of three classes, and it arrived at the configuration space whose topology fixes the statistics of particles. The witness shows the step that matters. Its core octahedron alone is planar; the leaves, which are the decoration that gives the reader its direction, are exactly what make the graph non-planar and pin the exchange to ±1.

> **Keyframe 13. (the exchange is the orientation character of a surface)** The two-token complex of $K_{3,3}$ has 15 configurations, 36 moves and 18 commuting squares, and every vertex link is a single cycle — a hexagon where both tokens sit on one side, a square where they are adjacent — so the complex is a closed surface. It is non-orientable, with Euler characteristic −3, hence of genus 5. The exchange reverses orientation: its mod-2 class is Poincaré dual to the first Stiefel–Whitney class w_1, the mod-2 cup form on the surface is non-degenerate of rank 5, and $w_1(z) = 1$. *Status.* Computed; the cup-product data agree with an independent run [18, §74].

5.4 More tokens

The exchange bit does not depend on how many particles there are. With k tokens the discretized model is the configuration space once every edge is subdivided into $k - 1$ pieces [19, 20]. On that model the first homology is the same for $k = 2$, 3 and 4 on every graph tested (Table 4), and on the non-planar ones the exchange, with the other tokens parked, is still the element of order two — in agreement with the theorem that for 2-connected graphs the first homology stops changing with the number of particles [21, Theorem 5]. K_5 with four tokens reproduces the worked example of Ko and Park, $H_1(B_4K_5) \cong \mathbb{Z}^6 \oplus \mathbb{Z}_2$ [22]. Without the subdivision the discrete model can give the wrong answer — $K_{3,3}$ with three tokens and no subdivision gives $\mathbb{Z}^5$ — a reminder that an arena has to be the space it stands for.

graph	$k = 2$	$k = 3$	$k = 4$	$k = 3$, unsubdivided
K_4 (planar)	$\mathbb{Z}^4$	$\mathbb{Z}^4$	$\mathbb{Z}^4$	$\mathbb{Z}^3$
cube (planar)	$\mathbb{Z}^6$	$\mathbb{Z}^6$	—	$\mathbb{Z}^6$
$K_{3,3}$	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$	$\mathbb{Z}^4 \oplus \mathbb{Z}/2$	$\mathbb{Z}^5$
K_5	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$	$\mathbb{Z}^6 \oplus \mathbb{Z}/2$
witness	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$	—	$\mathbb{Z}^{19} \oplus \mathbb{Z}/2$

Table 4. First homology with k tokens, each edge subdivided into k – 1 pieces (k = 3: into 2; k = 4: into 3). The last column omits the subdivision.

5.5 The residue: one circulation survives every square

Declare every 4-cycle of the arena instead of the commuting squares alone. The exchange dies: a spectator square and a rotation square on the same 4-cycle of the graph have the same trace and opposite swaps, so declaring both kills z. And of the 181 free directions at $n = 4$ exactly one survives. It survives at every size.

> **Keyframe 14. (the residue)** For every n the first rational cohomology of the arena with all its 4-cycles filled is one-dimensional: H^1(square-filled arena of Γ_n; $\mathbb{Q}$) $\cong \mathbb{Q}$. It is spanned by the pull-back of an integral cocycle f on the graph: $f = +1$ on a core edge from class i to class $i + 1$; from a leaf, 3 to its own core point and 1 to each point of the next class; 0 from a mark vertex to its corner; and $f(t_a \to t_b) = f(a \to b)$ on a mark triangle. The cocycle f vanishes on every 4-cycle of the graph and is ±3 on every triangle.

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Status. Proved: every arena 4-cycle either traces nothing, and is then a commuting square, or traces one 4-cycle of the graph, and every 4-cycle of the graph is traced; by Keyframe 12 this turns the question into one about the graph with its 4-cycles filled, whose first homology is spanned by a single triangle. Computed at arena level for $n \le 4$ and at graph level for $n \le 8$ [18, §§72, 74, 75].

The residue is the forward bias around the three classes, $0 \to 1 \to 2 \to 0$. It belongs to the object rather than the reader: no group is needed to define it, and the reader acts on it only by +1. Of all the circulations the graph carries, it is the one that every square relation lets through.

5.6 The clock: the residue read modulo 3

Reduce f modulo 3. On every edge $f(u \to v) \equiv \mathrm{cl}(v) - \mathrm{cl}(u)$, where cl is the class number of a vertex and a leaf or a mark vertex takes the class of its own core point. So modulo 3 the residue is exact, and its potential is the class number. On the arena the potential is $F(\{u, v\}) = 2(\mathrm{cl}\ u + \mathrm{cl}\ v) \bmod 3$. It is single-valued — and it is not invariant, because a reader element g that rotates the classes shifts it:

$$F(g \cdot x) \;=\; F(x) + \mathrm{rot}(g) \quad (\mathrm{mod}\ 3).$$

Keyframe 15. (the clock is the residue at another coefficient ring) The rational residue and the reader's order-3 clock are one integral cocycle read over $\mathbb{Q}$ and over $\mathbb{Z}/3$. The clock is the reader-relative class: the character $G \to \mathbb{Z}/3$ by which class rotations shift F. Its kernel on the witness is the reader of order 8, whose three classes on the interior are the three phases of the clock. It survives exactly when no class rotation fixes a pair of token positions. *Status.* Proved; computed at $n = 2, 3, 4$ [5; 18, §§68–71, 74].

An earlier lemma of the programme says the same in the language of periods. When every circulation of a class lies in $d\mathbb{Z}$, the class reduced modulo d is exact, so a single-valued clock always exists, and what is missing is exactly the lift of that clock from $\mathbb{Z}/d$ back to $\mathbb{Z}$ [16]. The hidden potential is not absent. It is present, modulo d. The base keeps d histories as one visible endpoint, and d is the number.

6. Price, drive, and the gap we know

6.1 A static potential drives nothing

A *price* is the difference of a potential, $p = \mathrm{d}\varphi$. Paper IV showed that an invariant price is constant on a reader's classes, so it is silent on every move inside a class; Paper V opened the moves where it is silent and found the methods of §4 there [4, 5]. Silence is not the only limit. If motion is driven by a price alone the system relaxes and stops: on the cycle arenas tested, a static potential produces no current at all, and adding a potential to any other drive changes the current through the residue channel by exactly nothing [15]. The potential is not weak at driving the channel. It is invisible to it.

6.2 Drives, and what a relation rule lets through

Write a rate law on the arena, with forward and backward rates $k(u \to v)$ and $k(v \to u)$, and let $a = \log(k(u \to v)/k(v \to u))$ be its antisymmetric part. By Kolmogorov's criterion the dynamics satisfies detailed balance — it settles into a stationary state with no current — exactly when a is exact, $a = -\beta\ \mathrm{d}E$ [25]. Everything else is a *drive*, measured by its circulation around loops: the cycle affinities of Schnakenberg's network theory [26]. On the witness, with rates that respect the reader, the count is the following.

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reader-invariant directions of the antisymmetric part a	on the witness
all of them (one per orbit of moves)	22
prices: exact, the equilibrium landscapes	7
drives: not exact, the cycle affinities	15
drives with zero affinity round every commuting square	2
… and round every 4-cycle	1 — the residue
… and round every triangle as well	0

Table 5. Prices and drives on the witness [18, Part I and §75]. Zero affinity round the commuting squares is what independence means: each token's rates do not depend on where the other token sits. The drives that remain are then the one-token drives, through the trace of Keyframe 12.

So the relation rules of §5 are statements about which loops a dynamics can drive. Independent particles admit only one-token drives, the invariant cycles of the graph. Balance around every 4-cycle leaves one drive, the residue of Keyframe 14: a steady push forward around the three classes. Balance around every short loop leaves none.

> **Keyframe 16. (the exchange cannot be driven, only phased)** Under any drive that respects independence the exchange has zero affinity: twice the exchange is a boundary, so $\oint_z a = ½ \oint_{\partial B} a = 0$. Real rates cannot pump a persistent swap. Only a phase sees it, through $\mathrm{Hom}(\mathbb{Z}/2, \mathrm{U}(1)) = \{\pm 1\}$. *Status.* Proved.

The three kinds of structure are now separate and each has its own instrument. Prices are equilibrium landscapes and are read by potentials. Drives are circulations and are selected by relation rules. The exchange is an integral class of order two, invisible to every real-valued rate and visible to a phase.

6.3 The gap, sized

What the topology does not choose is the symmetric part of the rates — the mobility $\kappa_{uv} = \kappa_{vu}$, one number per orbit of moves, 22 on the witness — and the scale β that turns a price into a rate ratio. Local detailed balance makes the split explicit:

$$k(u \to v) \;=\; \kappa_{uv} \cdot \exp(-\beta\, p_{uv} / 2), \qquad \text{so} \qquad k(u \to v) \,/\, k(v \to u) \;=\; \exp(-\beta\, p_{uv}).$$

It is a candidate closure, not a derivation: it fixes the ratio from the price and leaves the speed to κ. On a move of zero price the forward and backward rates are equal but need not be zero — a move can be unpriced and still be fast. One structural consequence is already forced. A demand for sustained winding round a residue cannot be the equilibrium energy of a closed system: on the cover where the winding becomes single-valued, such an objective is unbounded below in one direction, so a persistent demand for winding is necessarily open, driven, finite in horizon, or tied to a task [15].

This is the gap the programme knows about, and it is the good kind. It is located: the symmetric part of the rate law. It is sized: 22 mobilities and one scale on the witness. And it is measurable: forward and backward rates, the stationary current that a drive produces, and the ratio test $\log(k(u \to v)/k(v \to u)) = -\beta(E_v - E_u)$. We know how wide it is and what kind of measurement crosses it.

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7. Retained organization: the past does not have to keep running

7.1 The photograph, again

Return to the photograph of §1.2. Three things are true of it at once. It *exists* as an artifact. Its organization can *support* further interactions. And a particular encounter *selects* some of its distinctions and produces a result. It can exist without being examined; its potential does not need to be continuously activated; and its next use need not reproduce the event. The original event and the retained artifact occupy different continuations: the artifact is produced by one history and can enter another.

None of this is a picture laid over reality by a framework. The photograph exists, the recognition happens, the decision follows. NEXUS does not grant those events reality by naming them; it tries to describe what they do.

7.2 Learning moves the work

Let a system have a working state x, a retained organization m, and an incoming encounter e, and let one joint rule U update both:

$$(x_{t+1}, m_{t+1}) = U(x_t, m_t, e_t), \qquad T_m(x, e) = \mathrm{pr}_x\, U(x, m, e).$$

The joint rule never changes. The *effective* rule T_m — how this system now handles that encounter — changes whenever m does, so the same event can produce a different response before and after: $T_{m_0}(x, e) \neq T_{m_1}(x, e)$. No new law is needed for learning; the state was larger than it looked. Continuation can change the structure that determines continuation.

Before learning, the next move is selected by search. After it, a recognized situation activates a retained organization and the move is executed. The work of finding a continuation has become structure that supplies the continuation. That relocates cost rather than abolishing it, and an accounting that keeps the parts apart is

$$C_{\text{total}}(N) = C_{\text{acquire}} + C_{\text{maintain}}(N) + N \cdot C_{\text{run}},$$

under which retention pays when acquisition lowers the recurring cost enough over N encounters. A boat coming up onto the plane is the mechanical picture: still powered, still meeting the water, but in a different regime of continuation. Cumulative culture is the same thing at the scale of generations: a person learns a method without repeating the centuries that produced it, and the method becomes a component of work its discoverers could not have reached. Retained organization keeps the full cost of the past from becoming a recurring entrance fee to the future. The compression that matters here is compression of the reconstruction process, not of bits: a large representation can save enormous work, and a small one is useless if using it means rediscovering the result.

7.3 The mathematics already does it

Every keyframe of §§3–5 is, among other things, a photograph: a retained organization that is independent of the history that produced it and sufficient for the next use (Table 6).

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what is retained	what it is enough for	keyframe
one bit: the last method, or the second predecessor	choosing continue or turn as a function	9
length mod 3 and three turn parities	the endpoint of any walk	8
trace and swap	the class of any two-token loop	12
the clock modulo d	which of d sheets a history ends on	15
the reader cut, $F(S)$ states	continuing the addition from any set of read positions	5
seen bits, with the fibre held	the whole content, $\log_2 N!$	3

Table 6. Retained organizations already proved or computed in the programme. Each forgets something definite: the order of moves, the lift of the clock, the content of the fibre.

In each row the past has stopped running and remains operative. Each row also says exactly what the retained organization does not carry — the order of the moves, the non-abelian braid, the lift of the clock, the content of the fibre — and that is what makes it usable. A photograph that tried to keep everything would be the event again.

7.4 Reacting and reorganizing

Two kinds of execution need to be kept apart. *Reactive* execution propagates a change through an existing organization: an input arrives and the dependent responses follow, as in reactive programming. *Reflective* execution also changes the organization that will handle the next input. Looking at the photograph and recognizing the uncle revises what the next look will find:

$$e_t = \mathrm{EMMIT}(r, m_t, c_t), \qquad m_{t+1} = \mathrm{UPDATE}(m_t, e_t, c_t), \qquad e_{t+1} = \mathrm{EMMIT}(r, m_{t+1}, c_{t+1}).$$

The artifact r has not changed; the organization m that encounters it has, so the same artifact can emit something new. EMMIT need not create information from nothing — the new relation comes from combining distinctions in the artifact with distinctions the reader already holds. Not every reaction reorganizes, but where one does, the boundary between the two kinds is measurable in any computational system, which makes it the natural first test of a retained-continuation law (§9.4).

7.5 Pull, and realization

A constraint structure delimits a space of continuations; actual interactions make one of them concrete; and the realization changes what is available next:

$$C \to \mathcal{P}(C) \to R \to C'.$$

Baseball shows each arrow. Its governing relations — outs, innings, bases, the conditions under which one action changes the state of the game — came before every interface that carries them. The field, the board game and the Atari 2600 cartridge each realized enough of those relations to remain baseball while changing the motions completely, and each then became an environment in which people learned, played and built further games. The cartridge was pulled into existence by rules that already existed, and the timing matters: the rules precede the interface that realizes them. Once it existed it changed what could happen next. The direction can also reverse — a physical event is sensed and encoded, a computed design drives a machine, and the result becomes input again — and neither the analog nor the digital form is terminal.

Each such correspondence needs a contract that names its states, actions and observations, the map, what the map preserves, what it forgets, and its scope. Trace and swap is a contract of exactly this kind: the map

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from loops to (trace, swap) preserves every homology class and forgets the order of moves and every rate. An equivalence is always relative to what it promises to keep, and naming the kernel is what lets it travel between realizations without being stretched beyond them.

8. The incomplete universe

The pieces now fit together. The structure is complete. The catalogue of admissible moves on every arena of this paper is finite and fully written; the first homology of two tokens is known for every size; the methods of a walker are two; the residue is one. The execution is not complete. Walks continue, loops are traversed again, and what a walk keeps of its past becomes part of how it handles the next step. A result finished locally stays active globally. That is the sense of the title, and it puts no hole in the field. The gap is always the reader's, and every keyframe above was found by changing the reader — a coarser group, one more remembered step, a different relation rule, another coefficient ring — rather than by adding anything to the object.

We are not outside this. A game cartridge and a planet are very different realizations, but neither stands outside the constraints it runs on, and neither do we. The readers of this paper are subsystems of the systems they read, and there is no exterior from which a complete reading could be taken. That is not a limitation to regret. It is the reason the running continues — a universe that one of its parts could read completely would leave its parts nothing to do — and the reason every finished result becomes a starting point. The past is retained as organization; the organization is met by new conditions; the meeting produces what did not exist before; and that becomes organization in turn.

The optimism of this paper is specific. Starting from one statement and two prohibitions, the programme has built objects that keep landing on established mathematics and physics: the census of strongly connected digraphs, Frobenius reciprocity, Myhill–Nerode minimality, Abrams' configuration spaces, the braid groups of graphs, the exchange statistics of particles on networks. Each landing is a confirmation the construction could not have arranged for itself. The claims that were killed on the way (§11) were wrong turns caught by the instruments, and each one sharpened the road. The one gap that remains is named, located, sized and measurable. With the keyframes in place, the route between them can be drawn by anyone who builds a realization that respects them.

9. Where it is going

These are the keyframes for the next stretch, each with what would settle it.

9.1 Constitutive selection

Measure the mobilities and the scale on a physical realization of an arena. Test the ratio law $\log(k(u \to v)/k(v \to u)) = -\beta(E_v - E_u)$ against independently measured energy differences. Drive the residue and measure the current that circulates around the class triangles. Which constitutive law the world uses is an experimental question, and the topology has already said which numbers decide it.

9.2 The exchange in matter

Any network on which two indistinguishable particles hop realizes an arena. The prediction is classical and sharp: on a 3-connected non-planar network the exchange phase is ±1; on a planar one it is free [21]. The witness itself is a twelve-site design whose planar core is made non-planar by the same leaves that

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give its reader a direction, and the scaled family supplies larger designs whose free part is known in closed form, $12n^2 - 3n + 1$.

9.3 The integral side

Two integral data are now in hand that no rational reader can see: the order-3 clock character, which exists modulo d and not over $\mathbb{Q}$, and the order-2 exchange, which vanishes over $\mathbb{Q}$. On the witness the rational side of the relation-generation problem is closed — 15 orbit generators, attained [4] — and that says nothing about the integral side, where the relation gap problem and Wall's D(2) problem live. These two data are the first concrete footholds there.

9.4 Retained continuation as a theorem

When can a history be compressed into an organization that is sufficient for all future transitions, and what is the smallest such organization? The base cases are proved: one bit for the turn, the reader cut for the adder, trace and swap for loops of two tokens. The next case is a system whose retained organization changes its own effective rule (§7.2), where the boundary between reacting and reorganizing can be measured and a candidate law — the result of resolving a continuation is kept as organization that changes the accessibility of later continuations — can be tested against a system that could fail it.

9.5 The inverse problem

Given an observed sequence $a(n) = \pi(T^n(x_0))$, recover the smallest (X, T, π) that generates it. For π in base 16 the read-head is in hand (§3.4). For SHA-256 the honest state is also a keyframe. One reverse round is a closed-form computation; the true message word scores exactly zero under a probe that reads only carry geometry; and when depth is added without an answer key, the measured collapse factor of the cold basin is 1.00 per round in the schedule-free model — depth tightens the basin precisely as much as the search for each new word widens it [14]. The difficulty is relocated, not destroyed, and we know where it lives; the message schedule, which couples the words, is the named next experiment. The hexadecimal transformation space — how a single source expands under every way of reading it — is being mapped in parallel.

9.6 The standing four

standing aim	where it stands	keyframes
1. how a transformation is done, not its values	the address action; π read as an angle; alignment buys cost, not possibility	6
2. shapes are methods	literal: the shape is the Cayley graph of the methods; every path is continue or turn; the methods form a device for logic	7–11
3. read the missing methods without searching	one character answers every channel; trace and swap classify every loop	4, 12
4. something real to shoot for	exchange statistics on networks; the constitutive gap; the integral side	13–16, §9

Table 7. The programme's four standing aims, located.

10. Status ledger

	statement	status	source
1	shapes from two prohibitions: 0, 1, 5, 83	computed, $n \le 4$; recovered	§2.2; [24]

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	statement	status	source
2	C1 on the cycle ⟺ Abrams' hypothesis	computed; recovered	[15]; [19]
3	seen + hidden = $\log_2 N!$; orientation is a holonomy	proved	[9, 12]
4	one character answers every channel; the most symmetric reader is blindest	computed; recovered	[4]; [27]
5	adder cut law; $b_1 = C(n, 2) + (n - 1)$; carry capacity	proved, all n; computed	[6–8]
6	reading at an address; alignment buys cost	measured earlier; digits re-run	[17]
7	the interior is the Cayley graph of the methods	computed	[5]
8	continue and turn are the only rules with one more step	computed	[18, §65]
9	one remembered bit is the minimal memory	proved	[18, §66]
10	the device; AND only where the reader is small	computed	[18, §67]
11	turn, clock and one interior at every size	proved, $n \geq 3$; computed n = 3, 4	[18, §§69–71]
12	trace and swap: $H_1 \cong H_1(\Gamma) \oplus \mathbb{Z}/2$	computed n = 2…6; every n via [21]	[18, §§74–75]
13	the exchange is the orientation character of the $K_{3,3}$ surface	computed	[18, §74]
14	the residue: $H^1 \cong \mathbb{Q}$ for every n	proved via [21]; computed	[18, §§72–75]
15	the clock is the residue modulo 3	proved; computed $n \leq 4$	[18, §§68–74]
16	prices and drives, 22 = 7 + 15, then 2, 1, 0; the exchange is phase-only	computed; proved	[18, Part I, §75]
—	the constitutive law: mobilities and scale	open; the deciding numbers named	§6.3

Table 8. Every keyframe, with its status and where it is on record.

11. What was corrected on the way

The programme keeps its killed claims as navigational markers; its audit papers record them in full. Six of them shaped the keyframes above, and each was settled by running something rather than by argument.

claim	what replaced it	how it was found
at n = 3 a graph gives turn or clock, not both	both, once the face marks have no centre; the failure was an artefact of a fixed centre	building the marks without centres
the trace map is an isomorphism	an exact sequence whose kernel is the exchange; an isomorphism over $\mathbb{Q}$	a review, re-run
the residue is a blind spot for odd cycles	omission from the declared family: arenas with no odd cycle carry it too	bipartite arenas of C_6 and C_8
an odd interior factor forbids a decomposition into digits	true in base 2 only: b = 3, L = 2 gives $9 = 3^2$	changing the base [7]
a walk's endpoint is its phase plus one parity	it needs the turn parity at each of three positions	all 8190 words
the deck group of the cover is inertia	the period group is $d\mathbb{Z}$; cost needs a functional; using a channel (3) and erasing it (4) differ	breadth-first search on the cover [15]

Table 9. Corrections that shaped the keyframes.

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12. Method, authorship and reproducibility

The programme is driven by Dean A. Kulik. Computation, verification and drafting were carried out with Claude (Anthropic) in long working sessions. Drafts and results were also reviewed in parallel sessions with other AI systems, and every checkable point in those reviews was re-run before it was accepted or rejected; where an independent run reported the same numbers, that is noted.

All computations are exact — integer, rational or finite-field arithmetic — and use the Python standard library, with NetworkX for planarity tests and SymPy for the symbolic integral. Every number in this paper is drawn from the programme's ledger [18], where each result is recorded with the script that produced it, its controls and its scope. The scripts for the results first reported here are `n4.py`, `n56.py`, `surface.py`, `graph4.py`, `kpoint.py`, `planar_contrast.py` and `keyframes_check.py`. This paper is released under CC BY-NC 4.0.

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